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Transfer Learning-Enhanced Multi-objective Predictive Control for Electric Vehicular Platoon

Yanhong Wu, Defeng He, Ji Li, *Member, IEEE*, Quan Zhou, *Member, IEEE*, Hongming Xu

***Abstract*—The growing adoption of connected and automated vehicles has led to significant advancements in intelligent transportation. The computational efficiency stands as a critical bottleneck in the commercialization of connected and automated vehicles. This paper proposes a novel transfer learning-enhanced multi-objective predictive control strategy to mitigate the adverse effects of computational burden for vehicular platoon. A data-driven platoon model is established with subspace identification to characterize the non-ideal driving behavior and complex powertrain structure of electric vehicles. To balance multi-objective conflicts among driving safety, driving comfort and energy economy, a multi-objective cost function incorporating the predictive sequence is designed. Then, a grey wolf optimizer is devised to guide the search process, with the goal of achieving globally optimal trade-offs. Here, the knowledge of vehicle in the source domain, such as data-driven model and controller hyperparameters, is transferred to vehicles in the target domain. Based on the knowledge, each vehicle in the target domain just should update the knowledge in terms of the individual dynamic characteristic and tasks. With in this mind, a transfer learning-enhanced multi-objective predictive control strategy is developed to enhance the performance of tasks in the target domain. Finally, a hardware-in-the-loop experiment platform with the Carmaker and the driving simulator is conducted. The experimental results demonstrate that our proposed transfer learning-enhanced multi-objective predictive control strategy could improve the 28.8% computational efficiency while ensuring the satisfied platoon tracking performance.**

***Index Terms*—Vehicular platoon, multi-objective optimization, data-driven model, transfer learning.**

I. Introduction

Connected and autonomous vehicles (CAVs) with advanced artificial intelligence technologies are revolutionizing transportation, improving road safety and efficiency. CAVs represent a cutting-edge advancement in the automotive industry, characterized by the integration of vehicular technologies with communication networks and computational systems [1], [2]. Despite significant advancements in the advanced algorithm, the existing vehicular onboard hardware fails to meet the computational demands of intricate tasks, especially for large-scale vehicle platoon [3]. This limitation impedes the full utilization of advanced control algorithms, thereby necessitating the development of a feasible solution for vehicular platoon.

The exploration of control algorithms is crucial for ensuring the safe and efficient driving of the electric vehicular platoon. Several control approaches, such as sliding mode control, robust control and model predictive control have attracted significant attention in many industrial applications [4], [5]. These traditional methods often rely on fixed models and parameters, limiting their adaptability in dynamic traffic environments. Advanced techniques like adaptive dynamic programming, reinforcement learning control offer more flexibility but still struggle with scalability and computational demands [6]. Different from the above approaches that require explicit mathematical models or huge dataset resources, the data-driven model predictive control (DMPC) method could generate a data-driven model in terms of the historical data [7], [8]. Furthermore, the DMPC method retains the stability inherent in model-based approaches while leveraging the generalization advantages of data-driven methods, thereby enabling robust performance for vehicular platoon. Afterwards, several extended DMPC approaches such as reinforcement learning and artificial neural network were developed to improve the generalization under different conditions [9]–[11]. It should be noted that these methods require processing a substantial amount of data, which imposes a significant computational burden.

Actually, the computational cost increases as the scale of the platoon expands. The above data-driven schemes are hard to implement in vehicular chips since the enormous computational burden. A high computational efficiency enables vehicle platoons to perform control optimization, thereby improving string stability and enhancing the safety of platoon. In contrast, low computational efficiency may cause delays in decision-making and control execution, which can amplify disturbance propagation, degrade inter-vehicle spacing accuracy, and even compromise real-time feasibility [12]. Therefore, several model compression techniques such as weight pruning, parameter quantization and model distillation have been proposed to reduce computational costs [13], [14]. These techniques maintain the performance of the data-driven model within a certain accuracy range while significantly reducing the number of parameters and computational complexity [15]. However, these methods may necessitate additional resources to execute the compression algorithms, thereby increasing the overall computational burden. Furthermore, an edge computing approach was employed to reduce computational costs

This work was supported by the National Natural Science Foundation of China under Grant 62503426, the Zhejiang Province Natural Science Foundation under Grant LQN26F030031, LD26F020004 and the Royal Society of United Kingdom under the Grant IES R3243278. (Corresponding author: Defeng He.)

Yanhong Wu and Defeng He are with College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. (e-mails: yanhongwu@tju.edu.cn; hdfzj@zjut.edu.cn.)

Ji Li and Hongming Xu are with the Department of Mechanical Engineering, University of Birmingham, Birmingham B15 2TT, U.K. (e-mails: J.Li.1@bham.ac.uk; h.m.xu@bham.ac.uk.)

Quan Zhou is with the School of Automotive Engineering, Tongji University, Shanghai, China. (e-mails: q_zhou@tongji.edu.cn;)

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for vehicle platoon [16], [17]. The principle behind edge computing technology is to offload computational tasks from the vehicles to nearby edge devices, such as roadside units or edge servers, which can perform data processing and decision-making in real-time. It should be pointed out that such this method relies on expensive edge servers and requires high communication resources, which can hinder the commercialization process of autonomous driving. Recently, the transfer learning method offers a feasible solution for reducing computational burden in industrial domains [9], [18], [19]. This approach allows leveraging the knowledge acquired from the source domain to enhance the convergence performance in a similar target domain. However, this approach has not yet received sufficient attention in studies of vehicle platoon. Therefore, it is necessary to develop an efficient data-driven predictive control strategy with transfer learning for mixed vehicular platoon (MVP).

In addition to ensuring the driving safety, it is essential to prioritize energy economics and driving comfort for MVP [20]–[22]. The objective of driving economic aims to reduce energy consumption, making electric vehicle more sustainable and cost-effective [23]. Accordingly, the driving comfort is dedicated to enhancing the stable driving experience via reducing abrupt acceleration and deceleration behaviors. It should be pointed out that ensuring driving safety often requires conservative driving strategies and redundant safety mechanisms, which may conflict with the goal of maximizing energy efficiency. The essence of this phenomenon can be encapsulated as a multi-objective conflict challenge. However, the model predictive control (MPC) with a quadratic programming solver would get trapped in local optima while addressing multi-objective optimization tasks. Therefore, several optimization schemes such as particle swarm optimization [24], genetic optimization [25] and ant colony optimization [26] have been applied to enhance the capability of MPC. However, these schemes require extensive parameter adjusting, it would cause slow convergence and increased computational demands. The grey wolf optimization (GWO) offers an available solution by emulating the encircling prey behavior of grey wolves [27]. The superiority of GWO is that it could maintain a balance between exploration and exploitation, thus effectively convergence to the global optimal solution. Note that the heuristic nature of GWO might introduce inherent variability during in the optimization process, leading to the suboptimal solution in actual applications. Addressing these challenges necessitates balancing the unique attributes of GWO with the precise requirements of MPC. With this in mind, how to provide an available multi-objective optimization scheme becomes a difficult task that needs to be tackled.

The research mentioned in MVP modelling focuses on complex neural network-based data-driven models. It poses a significant computational burden for vehicular onboard hardware. Furthermore, the existing multi-objective optimization algorithm gets trapped in the local while addressing multi-objective conflicts. All these factors motivate us to construct an efficient control scheme for MVP. The main contributions of this paper are summarized below.

i) A subspace-based data-driven MVP model is developed to alleviate the adverse effects of uncertain dynamics. This approach utilizes a reduced-dimensional representation of the system's state space, significantly enhancing its ability to predict future vehicle states under uncertainty. This modelling approach marks a key innovation by offering a robust platoon model that adapts to complex driving environments.
ii) A GWO-based multi-objective optimization method is designed to balance the conflicts among driving safety, driving comfort and energy economy. This optimization method simultaneously considers multiple objectives, adjusting for trade-offs between vehicle stability, ride quality, and battery consumption. Furthermore, it could offer a robust alternative to traditional single-objective optimization methods in real-time vehicular platoon.
iii) A transfer learning-enhanced data-driven multi-objective predictive control (TDMPC) strategy is proposed to reduce the calculation burden. By transferring knowledge from source domain vehicles to target vehicles, this approach minimizes retraining efforts, enabling faster convergence to optimal policies while improving computational efficiency for large-scale vehicular platoon.

This paper is structured as follows. Section II presents the mixed vehicular platoon model. Section III illustrates the TDMPC strategy. Field experiments are carried out in Section IV. A comprehensive analysis of the results is presented in Section V. Section VI concludes the paper.

Notations: $\|\cdot\|_{\amalg}$ denotes the Frobenius norm. The Moore-Penrose pseudo inverse of matrix H is denoted as $H^{\dagger}$. $\triangle$ is the error between the current vector and the reference vector. $|A|$ stands for the absolute of A. $\|\cdot\|^2$ represents the L_2 norm.

II. Vehicular Platoon modelling

The MVP comprises one human driving vehicle (HDV) and M CAVs, all of which are pure electric vehicles. Fig. 1 illustrates the powertrain structure of an electric vehicle. The battery module provides electrical energy, which is converted into mechanical power by the motor through the DC/AC inverter. This mechanical power is transmitted via the transmission to the wheels, propelling the vehicle forward. Final drive components distribute torque to the wheels, while resistive power represents opposing forces like friction [28]. Electrical connections (green) show the power flow from the battery to the motor, while mechanical connections (red) represent the transmission of torque from the motor to the wheels.

Fig. 1 illustrates the powertrain structure of each electric vehicle. The battery module provides electrical energy that is transformed into mechanical force by the motor. Then, the tires are propelled by the motor through the transmission, ultimately driving the electric vehicles forward.

To derive a concise model, several reasonable hypotheses are given: 1) The vehicle is traveling on a dry and flat road, with negligible longitudinal slip of the tires. 2) The vehicle body is symmetric and rigid [29]. Furthermore, the internal resistance, temperature variations and capacity degradation of

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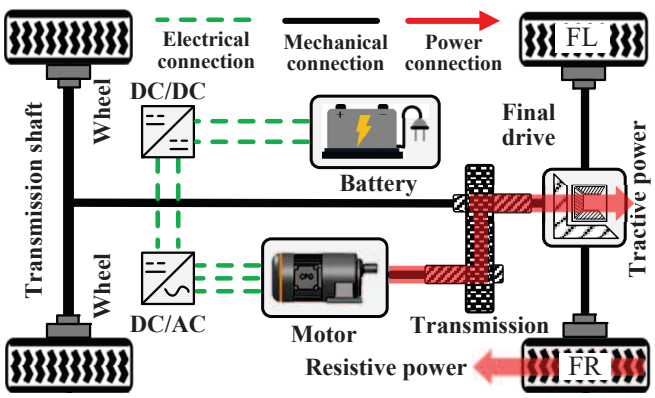

Fig. 1. Schematic diagram of pure electric vehicle.

the battery are ignored. Based on the powertrain properties of the electric vehicle [30], [31], one gets

$$\begin{cases} a = \dfrac{T_d i_g \eta_t}{mr} \\ \dot{\mathrm{SOC}} = -\dfrac{T_d}{K_e C_{bat}} \end{cases} \tag{1}$$

where SOC stands for the charge state of the battery. T_d denotes the demand torque. i_g represents the transmission ratio. η_t is the transmission efficiency. r stands for the tire radius. m denotes the mass of vehicle. a is the acceleration of the vehicle. C_{bat} represents the capacity of the battery. K_e stands for anti-electromotive force coefficient.

Then, a third-order vehicle kinematics model is utilized to describe the characteristics of the ith CAV ($i \in$[2,M]). Thus, we have

$$\begin{cases} \dot{h}_i = v_{i-1} - v_i \\ \dot{v}_i = a_i \\ \dot{a}_i = \frac{1}{\tau_i}(u_i - a_i) \end{cases} \tag{2}$$

where h_i denotes the spacing between adjacent vehicles. v_i is the velocity of the vehicle. τ_i stands for the unknown propulsion time delay. u_i represents the anticipated acceleration.

Furthermore, the optimal velocity model is employed to describe the characteristics of HDV [32]. Specifically,

$$\dot{a}_1 = \frac{1}{\tau_1}\Big(\alpha(V(h_1) - v_1) + \beta(v_0 - v_1) - a_1\Big) \tag{3}$$

where α and β are the headway gain and the velocity gain, respectively. $V(h_1)$ represents the spacing-dependent desired velocity function. Here, the virtual leading vehicle (VLV) is also a CAV with the third-order inertial model. As VLV sustains a velocity v_0, all following vehicles will converge to an equilibrium state (h^*, v^*). Note that v^* and h^* is determined by the condition $v^* = V(h^*)$ [33].

According to the properties of forward Euler, the mechanism model of HDV has a discrete state-space expression

$$x_1(k+1) = A_1 x_1(k) + E_1 x_0(k), \tag{4}$$

with matrices

$$A_1 = \begin{bmatrix} 1 & -T_s & 0 & 0 \\ 0 & 1 & T_s & 0 \\ \dfrac{\alpha_1 T_s}{\tau_1} & \dfrac{-(\alpha_1+\beta_1)T_s}{\tau_1} & \dfrac{(\tau_1 - T_s)}{\tau_1} & 0 \\ 0 & 0 & \dfrac{-m_1 R_1 T_s}{K_{e,1} C_{bat,1} i_g \eta_{t,1}} & 0 \end{bmatrix},$$

$$E_1 = \begin{bmatrix} 0 & T_s & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \dfrac{\beta_1 T_s}{\tau_1} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \; x_1(k) = \begin{bmatrix} \triangle h_1(k) \\ \triangle v_1(k) \\ a_1(k) \\ \triangle \mathrm{SOC}_1(\mathrm{k}) \end{bmatrix}$$

where $\triangle h_1(k) = h_1(k) - h^*$ and $\triangle v_1(k) = v_1(k) - v^*$. T_s denotes the sampling period. Note that x_0 and $x_1(k)$ represent the state vector of VLV and HDV, respectively. A_1 and E_1 are the system matrices of HDV.

Following similar principles with formula (4), the CAV could be deduced as

$$x_i(k+1) = A_i x_i(k) + B_i u_i(k) + E_i x_{i-1}(k) \tag{5}$$

with matrices

$$A_i = \begin{bmatrix} 1 & -T_s & 0 & 0 \\ 0 & 1 & T_s & 0 \\ 0 & 0 & T_s/\tau + 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; B_i = \begin{bmatrix} 0 \\ 0 \\ R_{g,i}\eta_{t,i} T_s/(m_i R_i I_i) \\ -T_s/(K_{e,i} C_{bat,i}) \end{bmatrix},$$

$$E_i = \begin{bmatrix} 0 & T_s & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \; x_i(k) = \begin{bmatrix} \triangle h_i(k) \\ \triangle v_i(k) \\ a_i(k) \\ \triangle \mathrm{SOC}_i(k) \end{bmatrix},$$

$u_i(k) = T_{d,i}(k)$ represents the input vector. A_i, B_i and E_i are the system matrices of the ith CAV. $x_i(k)$ denotes the state vector of the ith CAV.

Here, the predecessor-leader following (PLF) communication topology is utilized to the platoon [34]. Combining (4) and (5), the MVP system with one HDV and one CAV has the form

$$\begin{cases} \mathcal{X}_i(k+1) = \mathcal{A}_i \mathcal{X}_i(k) + \mathcal{B}_i \mathcal{U}_i(k) + \mathcal{E}_i \mathcal{W}_i(k) \\ \mathcal{Y}_i(k) = \mathcal{C}_i \mathcal{X}_i(k) \end{cases} \tag{6}$$

with matrices

$$\mathcal{A}_i = \begin{bmatrix} A_1 & 0 \\ E_i & A_i \end{bmatrix}, \; \mathcal{B}_i = \begin{bmatrix} 0 & 0 \\ 0 & B_i \end{bmatrix}, \; \mathcal{E}_i = \begin{bmatrix} E_1 \\ 0 \end{bmatrix},$$

$$\mathcal{U}_i(k) = \begin{bmatrix} 0 \\ u_i(k) \end{bmatrix}, \; \mathcal{X}_i(k) = \begin{bmatrix} x_1(k) \\ x_i(k) \end{bmatrix},$$

$\mathcal{C}$ denotes the identity matrix. $\mathcal{W}_i(k) = x_1(k)$ is the bounded disturbance.

III. TDMPC Strategy Design

In this part, the framework of TDMPC strategy with PLF communication topology is presented in Fig. 2. The predecessor leader following communication topology is applied to the platoon. Specifically, the data-driven model (7) is employed to construct the subspace predictor, which could generate the optimization predictive sequence. Then, a GWO-based multi-objective optimization scheme is developed to balance the conflicts among driving safety, driving comfort and energy economy. To reduce the calculation burden and enhance the generalization of the proposed strategy, a transfer learning approach is implemented to platoon control.

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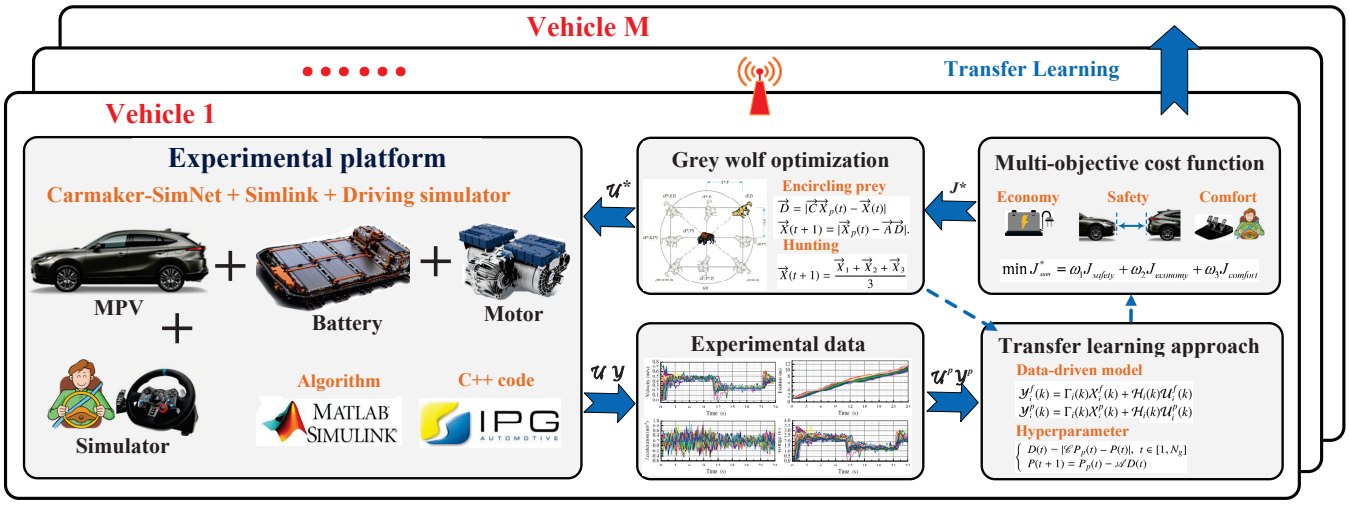

Fig. 2. The framework of the proposed TDMPC strategy.

A. Data-Driven Model Predictive Control

The DMPC offers several distinct advantages as an advanced control strategy in various industrial applications. However, the electric vehicular platoon involves complex powertrain structure and uncertain driver parameters. The subspace identification technique is presented to construct the DMPC [35].

Stacking up the collected input/output (I/O) trajectory from experiments, a data-driven model is expressed with Hankel matrices. That is

$$\mathcal{Y}_i^f(k) = \Gamma_i(k)\mathcal{X}_i^f(k) + \mathcal{H}_i(k)\mathcal{U}_i^f(k) + \mathcal{M}_i(k)\mathcal{W}_i^f(k) \tag{7a}$$

$$\mathcal{Y}_i^p(k) = \Gamma_i(k)\mathcal{X}_i^p(k) + \mathcal{H}_i(k)\mathcal{U}_i^p(k) + \mathcal{M}_i(k)\mathcal{W}_i^p(k) \tag{7b}$$

with matrices

$$\mathcal{Y}_i^f(k) = \begin{bmatrix} \mathcal{Y}_i(k+L) & \cdots & \mathcal{Y}_i(k+N) \\ \vdots & \ddots & \vdots \\ \mathcal{Y}_i(k+2L-1) & \cdots & \mathcal{Y}_i(k+N+L-1) \end{bmatrix},$$

$$\mathcal{Y}_i^p(k) = \begin{bmatrix} \mathcal{Y}_i(k) & \cdots & \mathcal{Y}_i(k+N-L) \\ \vdots & \ddots & \vdots \\ \mathcal{Y}_i(k+L-1) & \cdots & \mathcal{Y}_i(k+N-1) \end{bmatrix},$$

$$\mathcal{X}_i^p(k) = \begin{bmatrix} \mathcal{X}_i(k) \\ \vdots \\ \mathcal{X}_i(k+N-L) \end{bmatrix}^T, \mathcal{X}_i^f(k) = \begin{bmatrix} \mathcal{X}_i(k+L) \\ \vdots \\ \mathcal{X}_i(k+N) \end{bmatrix}^T,$$

$$\Gamma_i(k) = \begin{bmatrix} C_i(k) & C_i(k)\mathcal{A}_i(k) & \cdots & C_i(k)\mathcal{A}_i^{L-1}(k) \end{bmatrix}^T,$$

$$\mathcal{H}_i(k) = \begin{bmatrix} 0 & C_i(k)\mathcal{B}_i(k) & \cdots & C_i(k)\mathcal{A}_i^{L-2}(k)\mathcal{B}_i(k) \\ 0 & 0 & \cdots & C(k)\mathcal{A}_i^{L-3}(k)\mathcal{B}_i(k) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}^T,$$

$$\mathcal{M}_i(k) = \begin{bmatrix} I & C_i(k)\mathcal{E}_i(k) & \cdots & C_i(k)\mathcal{A}_i^{L-2}(k)\mathcal{E}_i(k) \\ 0 & I & \cdots & C_i(k)\mathcal{A}_i^{L-3}(k)\mathcal{E}_i(k) \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix}^T,$$

$\mathcal{U}_i^f(k)$, $\mathcal{U}_i^p(k)$, $\mathcal{W}_i^f(k)$ and $\mathcal{W}_i^p(k)$ have similar forms to $\mathcal{Y}_i^f(k)$, $\mathcal{Y}_i^p(k)$ and $\mathcal{M}_i(k)$. The subscripts p and f represent the past and future state of corresponding matrices. N and L denote the prediction horizon and excitive order.

Then, the data-driven model (7) is employed to formulate the linear subspace predictor equations. And the subspace linear predictor coefficients $L_w(k)$, $L_u(k)$ and $L_v(k)$ are derived by solving the following least squares problem

$$\min \left\| \mathcal{Y}_i^f(k) - [L_{w,i}(k)\ L_{u,i}(k)\ L_{v,i}(k)] \begin{bmatrix} \mathcal{Q}_i^p(k) \\ \mathcal{U}_i^f(k) \\ \mathcal{W}_i^f(k) \end{bmatrix} \right\|_{\Pi}^2 \tag{8}$$

where $\mathcal{Q}_i^p(k) = [\mathcal{Y}_i^p(k)\ \mathcal{U}_i^p(k)\ \mathcal{W}_i^p(k)]^T$ denotes the subspace prediction matrix corresponding to the past I/O trajectory.

According to the property of orthogonal projection, the optimal predicted sequences is deduced as

$$\hat{\mathcal{Y}}_i(k) = \mathcal{Y}_i^f(k) \begin{bmatrix} \mathcal{Q}_i^p(k) \\ \mathcal{U}_i^f(k) \\ \mathcal{W}_i^f(k) \end{bmatrix}^{\dagger} \begin{bmatrix} \mathcal{Q}_i^p(k)\ \mathcal{U}_i^f(k)\ \mathcal{W}_i^f(k) \end{bmatrix}^T \tag{9}$$

$$= \begin{bmatrix} L_{w,i}(k)\ L_{u,i}(k)\ L_{v,i}(k) \end{bmatrix} \begin{bmatrix} \mathcal{Q}_i^p(k)\ \mathcal{U}_i^f(k)\ \mathcal{W}_i^f(k) \end{bmatrix}^T.$$

Moreover, the MPC algorithm could offer several operational constraints to guarantee the platoon stability within the acceptable range [36].

B. Multi-Objective Optimization

It is crucial to ensure an efficient driving experience aligns with the driving safety and economic consideration [37], [38]. Therefore, a GWO-based multi-objective optimization method is constructed to trade-off the multi-objective conflicts. Note that each objective is described as follows. Driving safety J_s aims to keep the platoon spacing between ego vehicle and ahead vehicles. Driving comfort J_c is dedicated to minimizing the velocity and acceleration error among ego vehicle and ahead vehicles. Accordingly, the control input should also stay within a reasonable range. Energy economy J_e is dedicated to diminishing the deviation between the current SOC and its initial value.

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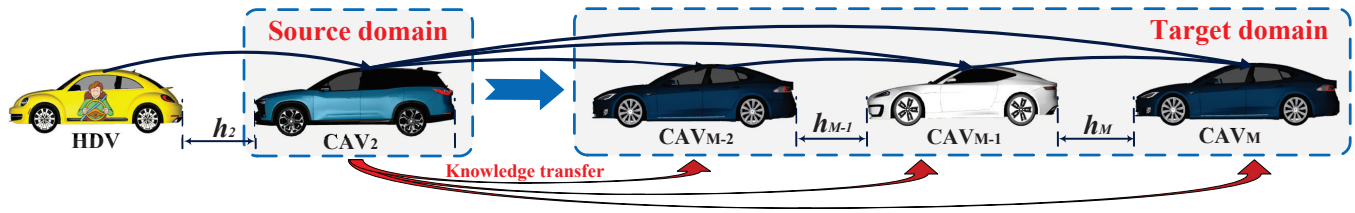

Fig. 3. The framework of the transfer learning for platoon.

The sum cost function involving driving safety, driving safety and energy economics is designed as

$$\begin{aligned} J_i^*(k) &= w_{s,i}J_{s,i}(k) + w_{c,i}J_{c,i}(k) + w_{e,i}J_{e,i}(k) \qquad (10)\\ &= \min \sum_{\tau=0}^{N-1} \Big(w_{s,i} \parallel \triangle h_i(k+\tau|k) \parallel^2 + w_{c,i}(\parallel \triangle v_i(k+\tau|k) \parallel_{Q_i}^2 \\ &\quad + \parallel a_i(k+\tau|k) \parallel_{R_i}^2 + \parallel T_{d,i}(k+\tau|k) \parallel_{F_i}^2) \\ &\quad + w_{e,i} \parallel \triangle \mathrm{SOC}_i(k+\tau|k) \parallel^2 \Big) \end{aligned}$$

where w_s, w_c and w_e denote the weight coefficients corresponding to three objectives with $w_s + w_c + w_e = 1$. Q, R and F are positive definite weight matrices.

In practice, the above multiple objectives often conflict or even get stuck in local optimal solutions during the solving process. GWO offers several advantages that align well with the structure and requirements of our multi-objective optimization formulation. Specifically, GWO's hierarchical leadership mechanism enables a natural balance between exploration and exploitation, which is essential for navigating the coupled and potentially conflicting objectives in platoon control. Therefore, the GWO is applied to address the global multi-objectives optimization issue [27].

This method emulates the encircling prey behavior of grey wolves, aiming to preserve a diverse spectrum of solutions. The solving process of GWO is modeled as

$$\begin{cases} D(t) = |\mathscr{B}P_p(t) - P(t)|, \ t \in [1, N_g] \\ P(t+1) = P_p(t) - \mathscr{A}D(t) \end{cases} \qquad (11)$$

where $P(t)$ and $P_p(t)$ denote the position of grey wolves and prey, respectively. N_g represents the maximum iteration step. $D(t)$ indicates the distance between wolf and prey. The vectors $\mathscr{A}$ and $\mathscr{B}$ are the coefficient vectors. Follow the above process, the optimal control input T_d is calculated and applied to drive MVP.

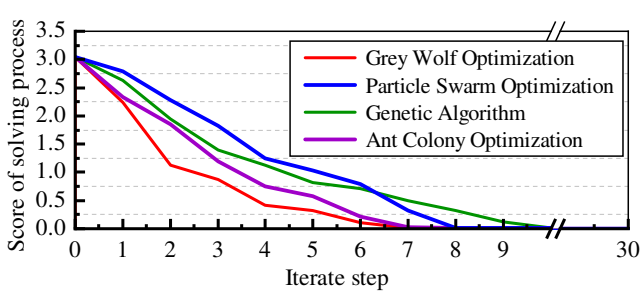

Fig. 4. The scores of different optimization approaches.

The performance scores of GWO, particle swarm optimization, genetic algorithm and ant colony optimization have been presented in Fig. 4. Compared to the other optimization approaches [39], the GWO demonstrates a significantly faster convergence, achieving optimal results within 6.5 iterations. It indicates that the proposed GWO exhibits superior effectiveness in solving multi-objective optimization problems.

C. Transfer Learning

For the distributed platoon control framework, each CAV requires establishing its own individual data-driven model based on the abundant dataset. It would result in a substantial computational burden, especially for large-scale vehicles under traffic system [18], [40]. Moreover, the behavior of HDV is hard to capture by following vehicles under the PLF communication topology. Transfer learning plays a significant role in improving computational efficiency, with its principles of feature sharing and model parameter transfer. This scheme involves applying learned knowledge in the source domain ($\mathcal{D}_S$) to the target domain ($\mathcal{D}_T$), thereby enhancing the performance of tasks in the target domain. Note that the source domain refers to the initial set of vehicle data, typically collected under well-defined conditions or configurations, used for training the model. The target domain refers to a different vehicle, where the learned knowledge is applied.

Algorithm 1 The Transfer Process of TDMPC

Input: Source domain dataset $\mathcal{D}_S$, target domain dataset $\mathcal{D}_T$, interation number N_t
Output: $L_w(k)$, $L_u(k)$, $L_v(k)$, $\mathscr{A}$ and $\mathscr{B}$ of the target domain
Initialize the data-driven model (7) and GWO hyperparameters (11) of the source domain
Acquire source knowledge using $\mathcal{D}_S$ (Vehicle $i = 2$)
Acquire target knowledge using $\mathcal{D}_T$ (Vehicle $i > 2$)
for $i = 1, \ldots, N_t$ **do**
    Transfer knowledge from the source domain to the target domain ($\mathcal{D}_S \rightarrow \mathcal{D}_T$)
    Calculate the information loss $\mathcal{D}_{KL}(P \parallel Q)$
    Update the data-driven model $\{L_w(k), L_u(k), L_v(k)\}$ and hyperparameters $\{\mathscr{A}, \mathscr{B}\}$ of the target domain
end for
return Optimal policy of the target domain

Here, the domain $\mathcal{D}_S$ consists of the feature space $\mathcal{S}_s$ and the marginal probability distribution function $\mathcal{P}(x)$.

$$\mathcal{D}_S = \{\mathcal{S}_s, \mathcal{P}(s)\}, s \in \mathcal{S}_s \qquad (12)$$

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TABLE I: EXPERIMENTAL PARAMETERS.

Symbol	Description	Value	Symbol	Description	Value
N	Prediction horizon	15	L	Excitive order	10
N_g	Maximum iteration step	30	w_s, w_c, w_e	Weight coefficients	0.5, 0.2, 0.3
T_s	Sampling period	$10Hz$	h^*	Desired spacing	100m
SOC_0	Initial SOC	70%	Q, R, F	Weight matrices	{20, 50, 60}*I_n
$\triangle s_{\max}$	Maximum spacing	110m	$\triangle s_{\min}$	Minimum spacing	90m
$\triangle v_{\max}$	Maximum velocity error	5m/s	$\triangle v_{\min}$	Minimum velocity error	-2m/s
$\triangle a_{\max}$	Maximum acceleration	10m/s^2	$\triangle a_{\min}$	Minimum acceleration	-10m/s^2

Accordingly, the domain $\mathcal{D}_T$ consists of the label space $\mathcal{S}_r$ andthe predictive function $\mathcal{F}(s)$. That is

$$\mathcal{D}_T = \{\mathcal{S}_r, \mathcal{F}(s)\}, s \in \mathcal{S}_r. \tag{13}$$

In practical applications, there are differences in the feature distributions of datasets between the source domain and the target domain, necessitating the classification and prediction of the unlabeled target domain dataset. Subsequently, the Kullback-Leibler divergence method is employed to measure the similarity of the feature distributions between the source domain and the target domain datasets [41]. This can be described by the following formula

$$\mathcal{D}_{KL}(P \parallel Q) = \sum_{s \in \mathcal{S}_r} P(s) \log \frac{P(s)}{Q(s)} \tag{14}$$

where $P(s)$ represents the probability distribution of the original unknown data. $Q(s)$ stands for the generated approximate probability distribution. $\mathcal{D}_{KL}(P \parallel Q)$ denotes the information loss from distribution $P(s)$ to distribution $Q(s)$.

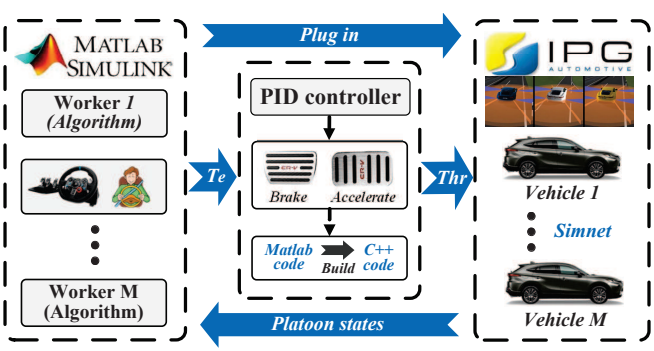

Fig. 5. Hardware framework of vehicular platoon platform.

Fig. 3 presents the flowchart of transfer learning for MVP. The knowledge of CAV_2 is regarded as the source domain including the pre-trained data-driven model and hyperparameters. Then, the knowledge is transferred to the following vehicles, namely target domain. Specifically, the collected dataset of CAV_2 is employed to train the data-driven model, and the hyperparameters of GWO-based controller also are obtained. Based on the knowledge, the following vehicles just should update this knowledge in terms of the individual dynamic characteristics and tasks. The details algorithm process has been described in Algorithm 1.

This approach enables subsequent vehicles to directly leverage the data-driven model—capturing both the complex dynamics of CAVs and the behavioral characteristics of HDVs without undergoing the full modeling procedure. In addition, it eliminates the need for the initial optimization stage required by the GWO-based controller, thereby improving the convergence speed of the overall platoon control strategy.

IV. Experiments Design

A hardware-in-the-loop experimental platform with a driving simulator and an IPG-CarMaker® software is developed. To evaluate the effectiveness of the proposed TDMPC strategy, several comparative experiments are conducted in this section.

TABLE II: VEHICLE PARAMETERS.

Vehicles	m(kg)	L_w(m)	C_{bat}(kWh)	r(m)	τ(s)
CAV_1(BYD)	2240	2920	77.8	0.35	0.12
CAV_2(NIO)	2250	2915	75.1	0.34	0.11
CAV_3(Tesla)	1610	2875	60.3	0.33	0.10
CAV_4(Toyota)	1890	2850	71.4	0.35	0.12

A. Experimental Platform

The hardware-in-the-loop experimental platform is depicted in Fig. 5. Specifically, all algorithms are deployed in Matlab language and execute a high-performance computer with Intel i7-14700KF (3.4GHz) CPU and RTX4070Ti GPU. It should be pointed out that vehicles can be controlled by the brake and acceleration, instead of the optimal control sequence (T_d). Therefore, a PID-based lower-level controller is designed, and it allows for tracking T_d by brake and acceleration of vehicles. Subsequently, all algorithms are compiled into C++ code and plugged into the IPG-CarMaker® software. To simulate mixed traffic scenarios, HDV is controlled by the driver with a driving simulator (see Fig.6). The real-time platoon information could be shared with the SimNet communication module.

In this experiment, the MVP with one HDV and three CAVs traveling on an 18km straight road, the desired inter-vehicle spacing is defined as 100m. Four default vehicle models including BYD (HDV), NIO (CAV_2), Tesla (CAV_3)

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Fig. 6. Driving simulator.

and Toyota (CAV_4) are chosen from IPG-CarMaker®. The reference velocity is determined by a high-speed driving cycle (Artemis Motorway) module. Furthermore, several key vehicle and experimental parameters have been listed in TABLE I and II.

B. Experimental Procedure

The experimental procedure is divided into two parts: collecting data and verifying strategies. A default control module of IPG-CarMaker® is chosen to steer the following vehicles. The collected results are utilized to train the data-driven model. Moreover, three experiments are conducted to validate the effectiveness of our proposed TDMPC strategy. The details of the experimental procedure are described as follows.

- **Exp1**. The mechanism-based MPC (MMPC) with uncertain dynamic parameters [7] is employed to the following vehicles.
- **Exp2**. The following vehicles are subjected to the data driven-based multi-objective predictive control (DMPC) strategy without the transfer learning method.
- **Exp3**. The proposed TDMPC strategy is applied to the following vehicles.
- **Exp4**. The meta learning-based data-driven multi-objective predictive control (MDMPC) strategy [42] is applied to following vehicles.

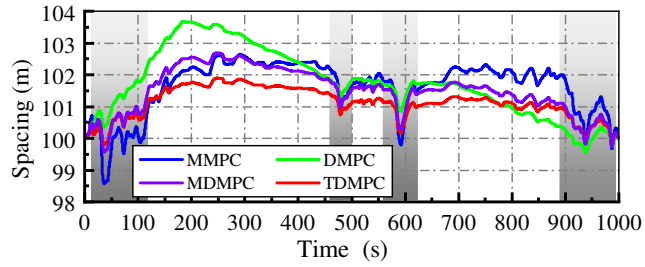
The collected experimental data $\mathbb{D} \in \mathbb{R}^{5\times 4}$ from 5 I/O trajectories $\{\triangle h_i(k), \triangle v_i(k), a_i(k), \mathrm{SOC}_i(k), u_i(k)\}$ of 4 vehicles. Note that this dataset is used to construct the data-driven model. Furthermore, different control strategies are embedded into the platoon, and similar data sets were collected to validate the effectiveness of the proposed algorithm.

V. Results and Discussions

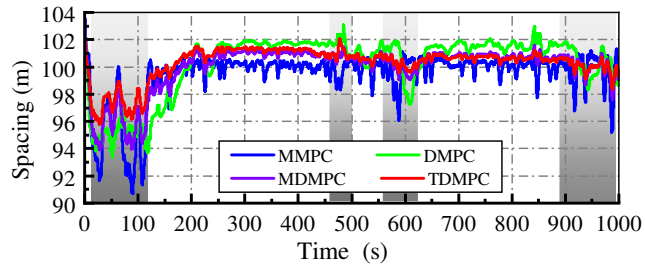
The platoon tracking performance and computational efficiency of different strategies are analyzed in this section.

A. Platoon Tracking Performance

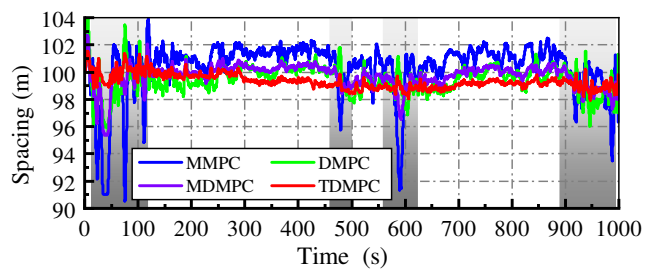
The platoon tracking performance is evaluated in terms of driving safety, comfort and economy. As depicted in Fig. 7,



(a) Spacing of CAV_2



(b) Spacing of CAV_3



(c) Spacing of CAV_4

Fig. 7. Results of the spacing with different strategies.

the platoon with the MMPC strategy performs the biggest spacing, and its peak reaches 90m. The DMPC and MDMPC strategies maintain more stable spacing, with MDMPC slightly outperforming DMPC in consistency. However, the TDMPC strategy provides the most stable and consistent spacing across all CAVs, with minimal fluctuations. Here, the confidence interval method has been applied to analyze the results of the spacing with different strategies. In Fig. 7(a), the 95% confidence intervals indicate distinct stability characteristics across the four methods: MMPC shows the largest variability with an interval of [101.606, 101.639]m, while TDMPC achieves the most consistent spacing with the narrowest interval of [101.104, 101.123]m. DMPC exhibits a wider interval of [101.709, 101.751]m, and MDMPC maintains relatively stable performance within [101.475, 101.503]m. Note that these vehicles (CAV_3 and CAV_4) are consistent across other vehicles as well. These results highlight that TDMPC provides the most reliable and consistent spacing control, while MMPC displays the greatest variation, suggesting potential for optimization.

The driving comfort of platoon is analyzed based on the results of velocity and acceleration. As illustrated in Fig. 8, the MMPC strategy exhibits significant velocity fluctuations,

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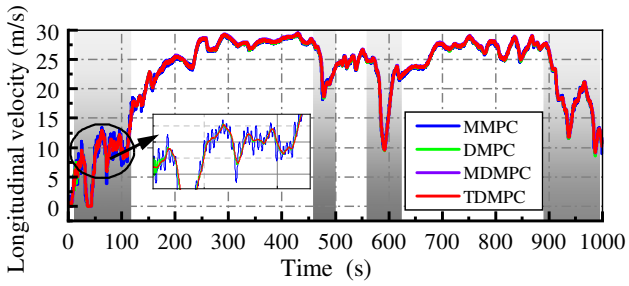
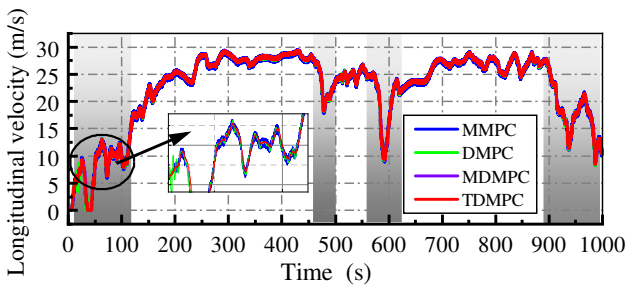
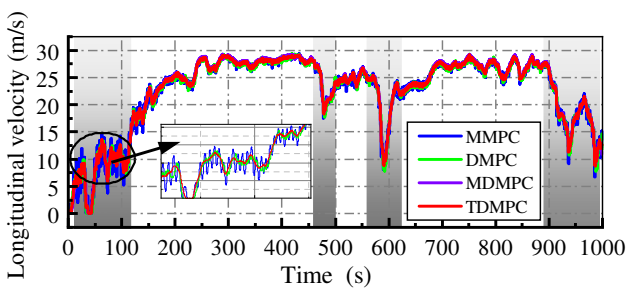


(a) Velocity of CAV_2



(b) Velocity of CAV_3



(c) Velocity of CAV_4

Fig. 8. Results of the velocity with different strategies.

leading to less stable driving, while DMPC and MDMPC strategies achieve smoother velocity tracking. Regarding acceleration, as depicted in Fig. 9, MMPC generates the highest acceleration values, followed by DMPC and MDMPC. In terms of interquartile range and median, TDMPC shows the lowest values, indicating the most consistent and smooth acceleration profile. These results suggest that TDMPC could provide superior velocity tracking and ensure the smoothest acceleration, offering the best driving comfort compared to the other approaches.

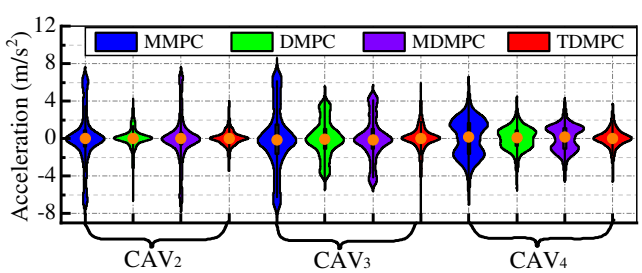
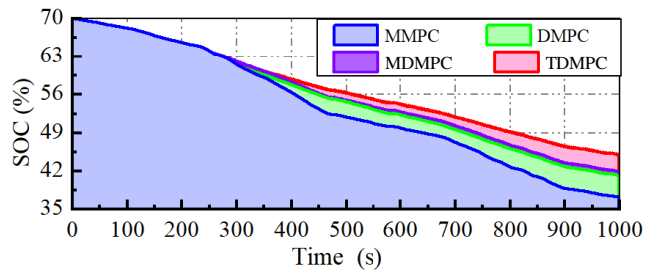
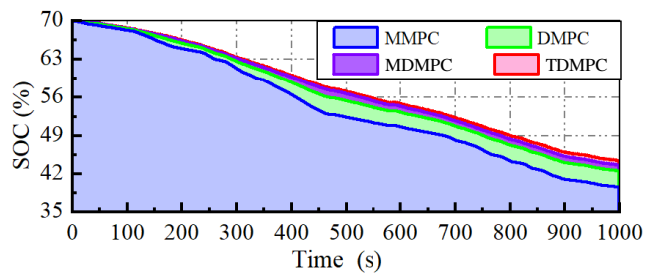


Fig. 9. Results of the acceleration with different strategies.

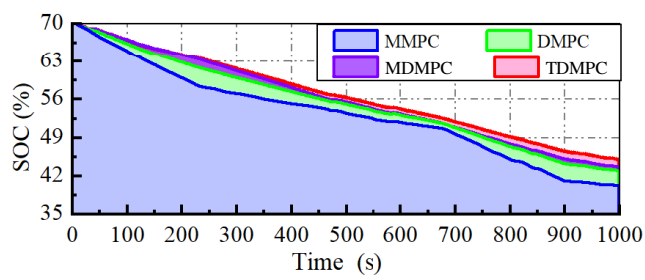
The energy economy aims to enhance battery efficiency by minimizing the consumption of SOC. The initial SOC is set at 70%, and it gradually diminishes as the vehicle's mileage increases. According to Fig. 10, the MMPC strategy exhibits the fastest decline rate compared to other strategies, with the final SOC of 36.2%, 38.5% and 41.7% for the following vehicles. In contrast, DMPC and MDMPC strategies exhibit similar and slower rates of SOC decline, indicating better energy management. TDMPC shows the slowest rate of SOC decrease, outperforming MMPC, DMPC, and MDMPC in terms of energy preservation. This suggests that TDMPC offers the most energy-efficient performance, making it the most effective strategy for reducing energy consumption in platoon driving.



(a) SOC of CAV_2



(b) SOC of CAV_3



(c) SOC of CAV_4

Fig. 10. Results of the SOC with different strategies.

To elaborate on the advantages of our proposed TDMPC strategy, the absolute maximum error (AMAXE) and root mean square error (RMSE) assessment indexes are introduced. Here, the average spacing error ($\triangle h$), velocity error ($\triangle v$), acceleration (a) and $\triangle$SOC of platoon are presented in TABLE III. The TDMPC strategy has the minimum AMAXE and RMSE, which implies that the desirable platoon stability could be ensured. Compared to MMPC, the TDMPC strategy shows a 35.9% reduction in $\triangle h$, 17.3% reduction in $\triangle v$, and 2.2% reduction in a, with a 2.8% improvement in SOC. Therefore,

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TABLE III: EXPERIMENTAL RESULTS ANALYSIS WITH AMAXE AND RMSE

Symbol	AMAXE				RMSE			
	$\triangle h$	$\triangle v$	a	$\triangle$SOC	$\triangle h$	$\triangle v$	a	SOC
MMPC	3.91m	5.25m/s	8.14m/s^2	0.048%	1.73m	0.32m/s	0.034m/s^2	5.47%
DMPC	2.53m	4.35m/s	7.96m/s^2	0.043%	1.56m	0.27m/s	0.025m/s^2	5.27%
MDMPC	2.32m	3.53m/s	6.92m/s^2	0.035%	1.42m	0.25m/s	0.023m/s^2	4.93%
TDMPC	2.14m	3.17 m/s	6.28m/s^2	0.031%	1.35m	0.22m/s	0.021m/s^2	4.77%

these results demonstrate that the proposed TDMPC strategy is superior to the MMPC while ensuring a similar performance as DMPC.

B. Computational Efficiency

Furthermore, the computational efficiency of the algorithm is crucial for MVP. Here, the convergence time of the solving process could determine the performance of computational efficiency. Note that CAV_3 and CAV_4 with TDMPC strategy execute superior convergence performance within 6 iterate steps than DMPC with 10 iterate steps and MDMPC with 7 iterate steps. This is because the knowledge of CAV_2 is transferred to CAV_3 and CAV_4, thereby eliminating the initialization process.

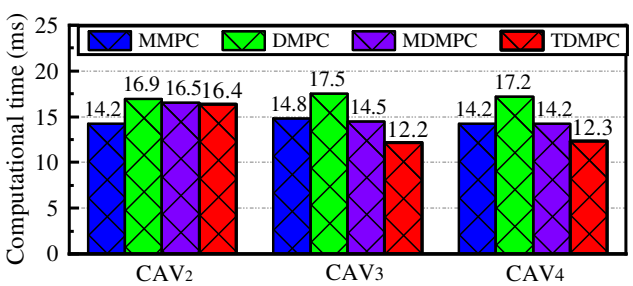

Fig. 11. Results of computational time.

Furthermore, the average computational time for each sampling interval among three strategies is recorded in Fig. 11. Following vehicles with the DMPC strategy required the maximum computational time, approximately 16.9ms, 17.5ms and 17.2ms, respectively. The proposed TDMPC strategy demonstrates shorter computation times than DMPC, MMPC and MDMPC, especially for CAV_3 and CAV_4 achieving 12.5ms and 12.2ms. This indicates a 24% improvement in solving efficiency with the TDMPC strategy. Given the limitations of computer performance, the MDMPC needs 1358, DMPC requires 1735s, while MMPC takes 1450s to complete a $100*10^3$ sampling simulation. Comparatively, the TDMPC strategy reduced this time to 1235s, thus saving 28.8% in computational time. In addition, the 95% confidence intervals reveal distinct differences in computational performance among the four control strategies. Specifically, MMPC and DMPC exhibit relatively wider intervals (e.g., CAV_3 with MMPC [14.55, 15.05]ms; DMPC [17.28, 17.72]ms), indicating greater variability in computation time. In contrast, MDMPC and TDMPC maintain consistently narrower ranges across all vehicles, such as TDMPC in CAV_3 with [12.05, 12.35]ms, demonstrating higher stability and lower computational burden. These trends are observed similarly among the three CAVs, suggesting that TDMPC achieves the most efficient and robust real-time computational performance among the compared approaches. The reason behind this phenomenon is that transfer learning method allows the following vehicles to obtain the data-driven model involving the complex vehicle dynamic model of CAV and driver behavior of HDV without the modelling process. Moreover, the initial optimization process of TDMPC could be bypassed, thereby enhancing the convergence speed of the platoon control strategy.

Therefore, it can be concluded that our proposed TDMPC strategy could improve the computational efficiency while ensuring the satisfied platoon tracking performance.

C. Discussion

The above experimental findings reveal several intriguing insights into the application of the proposed TDMPC strategy. Notably, the data-driven model could capture state transitions across different domains significantly to improve the learning efficiency, reducing the reliance on extensive data collection for each new scenario. However, an interesting observation is that while transfer learning accelerates convergence, there appears to be a sensitivity to vehicle-specific dynamics, suggesting that further refinement is needed for seamless adaptation in heterogeneous platoons. Additionally, the results indicate that while the data-driven model enhances overall performance, its capacity to generalize across highly variable traffic conditions requires deeper exploration, particularly in more complex, real-world settings. These findings underscore the need for a more nuanced understanding of model transfer and its limitations. In addition, the proposed strategy assumes a fixed communication topology and ideal V2X links without delays or packet losses. However, real-world wireless channels are subject to dynamic connectivity variations, stochastic latency, and transmission dropouts, which may influence both the stability and performance of platoon control. Incorporating these communication uncertainties into the control design will be an important direction for future research.

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VI. Conclusion

In this paper, we have developed a TDMPC strategy to address the issues of computational burden of MVP. The data-driven vehicular platoon model is constructed, it could alleviate the adverse effects caused by the non-ideal driving behavior and the dynamic uncertainty of the complex powertrain structure. Then, the GWO method has been developed to solve the optimal solutions amidst conflicting multi-objectives. Furthermore, the transfer learning method has been designed to share the knowledge within platoon, thereby reducing the computational burden. Finally, the experimental results demonstrate that our proposed TDMPC strategy could provide satisfactory platoon tracking performance and save 28.8% computational efficiency. In this paper, limited experimental scenarios and datasets make it challenging to validate the generalizability of the proposed strategy on complex scenarios. Therefore, exploring the application of our proposed strategy to actual vehicles will constitute a distinct research task. In future work, we will focus on testing the proposed strategy and investigating the effects of network-induced in actual vehicles.

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This article has been accepted for publication in IEEE Transactions on Transportation Electrification. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TTE.2026.3665598

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Yanhong Wu received the M.S. degree in vehicle engineering from Chongqing Jiaotong University, Chongqing, China, in 2020, and the Ph.D. degree with the School of Electrical Automation and Information Engineering, Tianjin University, Tianjin in 2024, China. He was a visiting Ph.D. with the University of Birmingham, U.K.

He is currently a Lecturer with the College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. His research interests include data-driven modelling, model predictive control, and their applications on connected and autonomous vehicles.

Defeng He received the B.S. degree in thermal energy and power engineering from Central South University, Changsha, China, in 2001, and the Ph.D. degree in control science and engineering from University of Science and Technology of China, Hefei, China, in 2008.

He is currently a Full Professor with the College of Information Engineering, Zhejiang University of Technology, Hangzhou, China. His research interests include model predictive control and its applications to connected autonomous systems.

Ji Li (Member, IEEE) received the Ph.D. degree in mechanical engineering from the University of Birmingham, U.K., in 2020.

He is currently an Assistant Professor and works on the Connected and Autonomous Systems for Electrified Vehicles (CASE-V) at the Birmingham CASE-V Automotive Research and Education Centre. His current research interests include computational intelligence, data-efficient modelling, feature selection, human-machine system, and their applications on CAVs.

Quan Zhou (M'17) received his BEng and MEng in vehicle engineering from Wuhan University of Technology, in 2012 and 2015, respectively. He received the Ph.D. in mechanical engineering from the University of Birmingham (UoB) in 2019. From 2021 to 2023, he was an Assistant Professor in Automotive Engineering and led the Research Group of Connected and Autonomous Systems for Electrified Vehicles (CASE-V) at UoB.

He is currently a Professor with School of Automotive Studies, Tongji University, Shanghai, China. His research interests include evolutionary computation, fuzzy logic, reinforcement learning, and their application in vehicular systems.

Hongming Xu received the Ph.D. degree in mechanical engineering from Imperial College London, London, U.K., in 1995.

He is a Professor of energy and automotive engineering at the University of Birmingham, Birmingham, U.K., and the Head of Vehicle and Engine Technology Research Centre. He has six years of industrial experience with Jaguar Land Rover, Coventry, U.K. He has authored and coauthored more than 500 journals and conference publications on advanced vehicle powertrain systems involving both experimental and modelling studies.

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